

Quiz — Projectile Motion at an Angle

Physics · Grade 11 · Multiple choice · 5 questions

Time: 15 min
Total: 5 marks

Name: Class: Date:

Data: $g = 10 \text{ m/s}^2$ | $\sin 30^\circ = 0.5$, $\cos 30^\circ = 0.866$ | $\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$ | $\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$
| air resistance is neglected.

Instructions: choose the **one** correct answer for each question and write its letter in the box.

Q1 A projectile is fired from the ground with a velocity of **40 m/s** at **30°** to the horizontal, as shown in Fig. 1. The total **time of flight T** is:

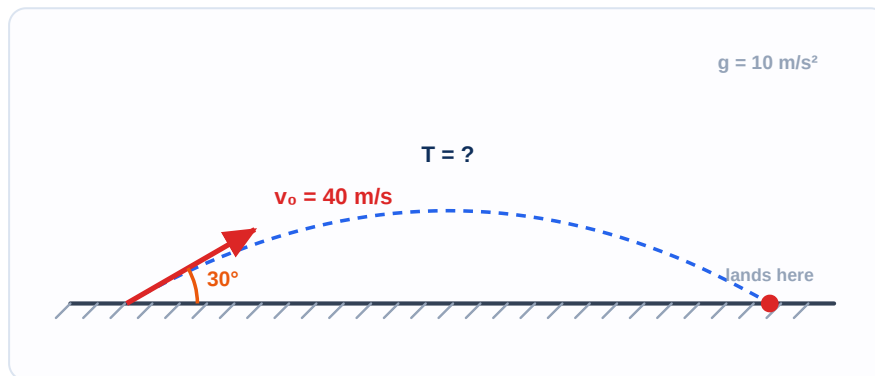


Fig. 1

A) 2 s

B) 4 s

C) 6 s

D) 8 s

Q2 A ball is projected with a velocity of **20 m/s** at **53°** to the horizontal (Fig. 2). The **maximum height H** reached by the ball is:

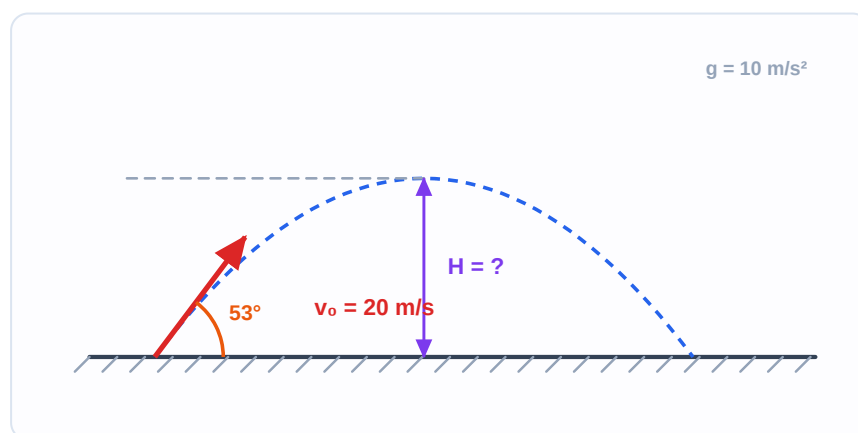


Fig. 2

A) 8 m

B) 12.8 m

C) 16 m

D) 20 m

Q3 Fig. 3 shows the two components of the initial velocity of a projectile. The magnitude of the initial velocity v_0 and the angle of projection θ are:

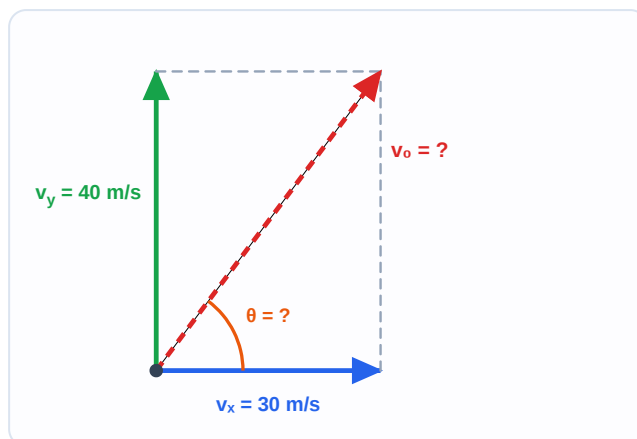


Fig. 3

A) 50 m/s and 37°

B) 50 m/s and 53°

C) 70 m/s and 53°

D) 35 m/s and 45°

Q4 A projectile is launched with a velocity of **50 m/s** at **37°** to the horizontal. Its **speed at the highest point P** of the path (Fig. 4) is:

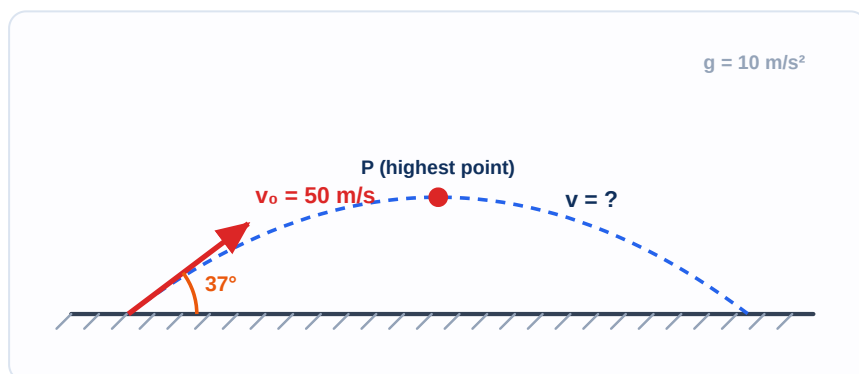


Fig. 4

A) zero

B) 30 m/s

C) 40 m/s

D) 50 m/s

Q5 Two balls A and B are projected from the same point with the **same speed**, at 30° and 60° to the horizontal (Fig. 5). Which statement is **correct**?

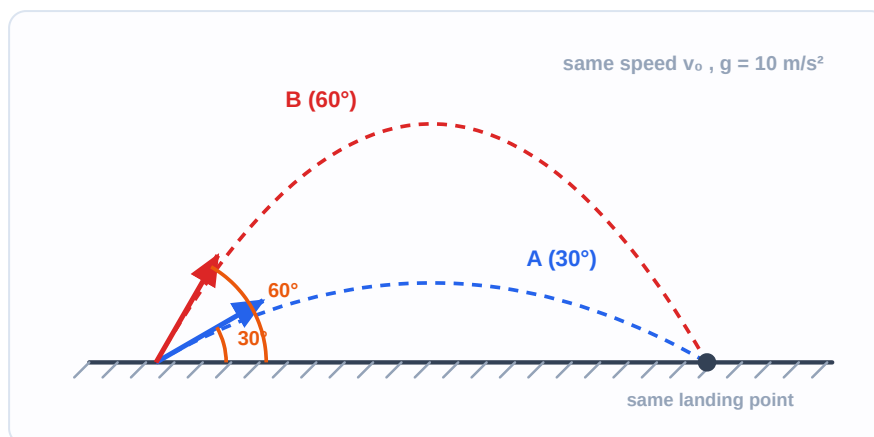


Fig. 5

A) They have the same range and the same maximum height.

B) They have the same range, but B rises higher and stays longer in the air.

C) A has a greater range than B.

D) B has a greater range and a shorter time of flight.

✓ Model answer — final answers

Q	Answer	Final value
1	B	$T = 4 \text{ s}$
2	B	$H = 12.8 \text{ m}$
3	B	$v_0 = 50 \text{ m/s}$, $\theta = 53^\circ$
4	C	$v = 40 \text{ m/s}$
5	B	same range ; B is higher and stays longer